1. **Inventory Management System**:
    - Create a system to manage inventory for a small business.
    - Include features like adding products, updating stock levels, and generating reports.

Project Overview

The Inventory Management System will allow users to manage products, update stock levels, and generate reports. The application will include features for adding, updating, deleting, and viewing products, as well as tracking inventory levels.

Project Structure

1. **Main Program**:
    - Entry point of the application.
    - Menu for navigating different functionalities.
2. **Product Class**:
    - Properties: ID, Name, Quantity, Price.
    - Methods: DisplayProductInfo().
3. **Inventory Class**:
    - List of Product objects.
    - Methods: AddProduct(), UpdateProduct(), DeleteProduct(), ViewProducts(), GenerateReport().

Detailed Steps

1. Setup and Initialization

- Create a new C# console application project.
- Define the Product class with properties and methods.
- Define the Inventory class to manage a list of products.
- Work with JSON file

2. Main Menu

- Implement a main menu with options for:
    - Adding a product.
    - Updating a product.
    - Deleting a product.
    - Viewing all products.
    - Generating a report.
    - Exiting the application.

3. Adding a Product

- Prompt the user for product details (ID, Name, Quantity, Price).
- Create a new Product object and add it to the inventory list.
- Confirm the addition to the user.

4. Updating a Product

- Prompt the user for the product ID to update.
- Search for the product in the inventory list.
- If found, prompt for new details and update the product.
- Confirm the update to the user.

5. Deleting a Product

- Prompt the user for the product ID to delete.
- Search for the product in the inventory list.
- If found, remove the product from the list.
- Confirm the deletion to the user.

6. Viewing Products

- Display all products in the inventory list.
- Include details like ID, Name, Quantity, and Price.

7. Generating a Report

- Calculate total inventory value.
- Display a summary of inventory levels.
- Optionally, save the report to a file.